Week 6: "Insight Extraction and Hypothesis Testing"

Dashboard Questions:-

The following questions were considered while creating the dashboard:-

1. What does the total number of accidents represent?
2. How did you calculate the total accidents from the data?
3. What is meant by severity and why is it important?
4. Why did you use average severity and what does it indicate?
5. How do weather conditions influence accidents?
6. Which time period shows the highest and lowest number of accidents?
7. Which city has the highest number of accidents and why?
8. How can you identify accident-prone regions from the data?
9. What is the purpose of using filters in your analysis?
10. What are the key findings from your overall analysis?

And the insights for the above mentioned questions are:-

1. It represents the total number of accident cases present in the dataset after applying filters.
2. It is calculated by counting the total number of records, usually using an ID or time field.
3. Severity shows how serious an accident is, and it helps in understanding the impact level of accidents.
4. Average severity gives the overall seriousness of accidents instead of just the count, making analysis more meaningful.
5. Weather conditions like rain and fog increase accidents due to low visibility and slippery roads.
6. The highest and lowest time periods indicate peak and low accident occurrences, which may depend on traffic and seasonal factors.
7. The city with the highest accidents usually has more population, vehicles, and traffic congestion.
8. Regions with higher accident counts are identified as accident-prone areas.
9. Filters help in analyzing specific data by narrowing down results based on conditions like city, time, or severity.
10. Key findings include:
    - Some cities have more accidents due to traffic density
    - Weather plays an important role in accidents
    - Accident numbers vary over time
    - Certain regions are more accident-prone